Northeastern University

Project Report – Multi-Object Detection & Localization (Milestone 2)

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Group 3

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Repository

GitHub Project Repository: https://github.com/rithika-sr/discriminative-deep-learning-project

All source code, training scripts, and detection outputs are available in the repository. This report focuses on **dataset generation, model training, results, analysis, and conclusions** for the multi-object detection and localization task.

1. Introduction

Building on the single-object classification system established in Milestone 1 (where MobileNetV2 achieved 98.94% test accuracy), Milestone 2 extends the project to **multi-object detection and localization**. The goal is to detect, identify, and locate multiple objects within a single composite image using the YOLOv8 object detection framework with transfer learning.

This milestone addresses a significantly more complex task: rather than classifying a single object per image, the model must simultaneously predict bounding box coordinates, class identities, and confidence scores for multiple objects in each image. Transfer learning from COCO-pretrained weights enables effective training despite a relatively small custom dataset.

2. Dataset Description

2.1 Multi-Object Dataset Generation

The multi-object dataset was synthetically generated by compositing individual object images from the Milestone 1 dataset onto a uniform canvas. A **zero-overlap** placement strategy was employed, ensuring that objects are clearly separated with no bounding box overlap. This approach produces cleaner annotations and more well-defined training examples for the detector.

Dataset Summary

- **Total composite images:** 400
- **Image resolution:** 640 × 640 pixels (YOLO standard)
- **Objects per image:** 1–5 (randomly selected; some images received fewer objects due to spacing constraints)
- **Total annotated instances:** ~1,200 bounding boxes
- **Object classes:** 35 (same as Milestone 1)
- **Background:** Light gray canvas (RGB: 240, 240, 240)
- **Annotation format:** YOLO `.txt` format (`class_id x_center y_center width height`, normalized to [0, 1])
- **Overlap policy:** Zero overlap — objects placed with guaranteed separation

Dataset Split

- **Training set:** 320 images (80%)
- **Validation set:** 40 images (10%)
- **Test set:** 40 images (10%)
- **Total:** 400 images

2.2 Image Generation Process

The composite image generation pipeline follows these steps:

1. **Random Class Selection:** 1–5 object classes are randomly sampled per image from the 35 available classes.
2. **Object Loading:** A random image is selected from the chosen class folder.
3. **Aspect-Preserving Resize:** Objects are resized to a maximum of 200 pixels per dimension while maintaining the original aspect ratio.
4. **Zero-Overlap Placement:** Objects are placed on the 640×640 canvas using a collision-detection algorithm that enforces **zero overlap** between bounding boxes. If an object cannot be placed without overlapping, the image receives fewer objects than initially targeted.
5. **YOLO Annotation:** Bounding box coordinates are automatically computed and saved in normalized YOLO format.

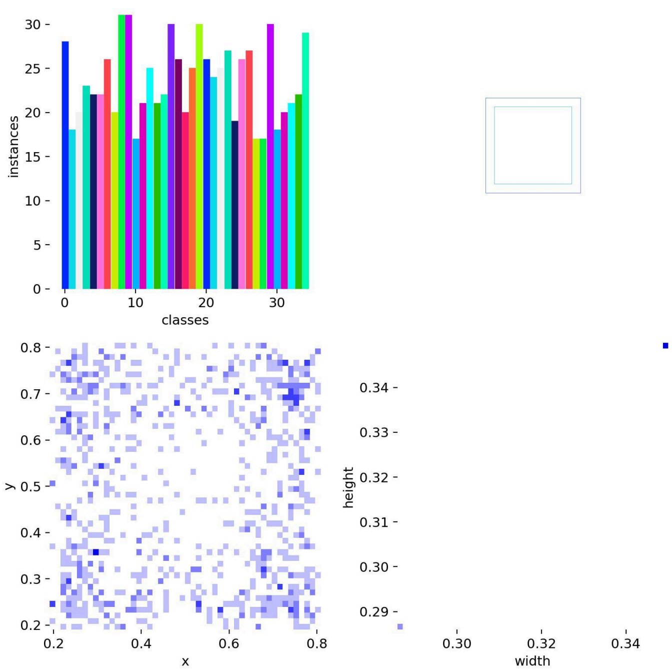
Note: 65 out of 400 images had reduced object counts due to spacing constraints from the zero-overlap policy. This trade-off ensures cleaner annotations at the cost of slightly fewer objects in some images.

Dataset Generation Output

```
======================================================================
GENERATING MULTI-OBJECT DATASET - ZERO OVERLAP
======================================================================

Generating 320 images for train set...
  Generated 20/320 images (3 objects in last image)
  Generated 40/320 images (2 objects in last image)
  Generated 60/320 images (2 objects in last image)
  Generated 80/320 images (3 objects in last image)
  Generated 100/320 images (3 objects in last image)
  Generated 120/320 images (3 objects in last image)
  Generated 140/320 images (4 objects in last image)
  Generated 160/320 images (2 objects in last image)
  Generated 180/320 images (3 objects in last image)
  Generated 200/320 images (2 objects in last image)
  Generated 220/320 images (1 objects in last image)
  Generated 240/320 images (2 objects in last image)
  Generated 260/320 images (3 objects in last image)
  Generated 280/320 images (2 objects in last image)
  Generated 300/320 images (3 objects in last image)
  Generated 320/320 images (2 objects in last image)
✓ Completed train set: 320 images

Generating 40 images for val set...
  Generated 20/40 images (1 objects in last image)
  Generated 40/40 images (4 objects in last image)
✓ Completed val set: 40 images

Generating 40 images for test set...
  Generated 20/40 images (2 objects in last image)
  Generated 40/40 images (4 objects in last image)
✓ Completed test set: 40 images

======================================================================
DATASET GENERATION COMPLETE - ZERO OVERLAP
======================================================================
Total images generated: 400
  Train: 320
  Val: 40
  Test: 40

⚠️ Note: 65 images had reduced objects due to spacing constraints

✓ Created dataset.yaml at: /Users/rithika/Documents/Discriminative_Project/milestone2/data/dataset.yaml
  Classes: 35

======================================================================
ALL DONE! ✓
======================================================================
```

2.3 Dataset Label Distribution

The training label distribution shows a relatively balanced spread of class instances across the dataset, with bounding box positions distributed across the canvas and consistent object sizes.



3. Model Architecture & Training Setup

3.1 YOLOv8s Architecture

YOLOv8s (small variant) was selected as the detection model, offering a strong balance between accuracy and computational efficiency.

Specification	Value
Architecture	YOLOv8s (small)
Input Resolution	640 × 640 pixels
Total Parameters	11,139,129
GFLOPs	28.5
Network Layers	73 (after fusion)
Detection Heads	3 scales (small, medium, large)

3.2 Transfer Learning Strategy

- **Pre-trained weights:** COCO dataset (80 classes)
- **Fine-tuned for:** 35 custom object classes (overriding `nc=80` with `nc=35`)
- **Trainable parameters:** 11.1M
- **Pre-trained model size:** 21.5MB (downloaded from Ultralytics)

3.3 Training Configuration

Parameter	Value	Parameter	Value
Base Model	`yolov8s.pt`	Early Stopping Patience	15 epochs
Epochs	100 (max)	Device	MPS
Batch Size	16	Workers	4
Image Size	640	Box Loss Gain	7.5
Optimizer	Adam	Classification Loss Gain	0.5
Learning Rate (initial)	0.001	DFL Loss Gain	1.5
Learning Rate (final)	0.01	Mosaic	1.0
IoU Threshold	0.7	Erasing	0.4
Momentum	0.937	Weight Decay	0.0005

3.4 Data Augmentation (Applied by YOLOv8)

YOLOv8 applies the following augmentations automatically during training:

- Mosaic augmentation (combines 4 training images)
- Random horizontal flipping (50% probability)
- HSV color space augmentation (hue: 0.015, saturation: 0.7, value: 0.4)
- Random erasing (40% probability)
- Image scaling (0.5) and translation (0.1)
- Random perspective and rotation (up to 12°)

Training Configuration Output

```
======================================================================
YOLO TRAINING SCRIPT - MILESTONE 2 (400 IMAGES)
======================================================================

✓ Dataset config found: /Users/rithika/Documents/Discriminative_Project/milestone2/data/dataset.yaml
✓ Number of classes: 35
✓ Training path: images/train
✓ Validation path: images/val

======================================================================
TRAINING CONFIGURATION
======================================================================
Model: yolov8s.pt
Epochs: 100
Batch size: 16
Image size: 640
Training images: 320
Validation images: 40
Test images: 40
Output directory: /Users/rithika/Documents/Discriminative_Project/milestone2/runs

======================================================================
LOADING YOLO MODEL
======================================================================
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolov8s.pt to 'yolov8s.pt': 100% ━━━━━━━━━━━━ 21.5MB 40.5MB/s 0.5s
✓ Loaded yolov8s.pt with pre-trained weights

======================================================================
STARTING TRAINING
======================================================================
Expected training time: 1-2 hours on M3 MacBook
Progress will be shown below...
Training will save checkpoints every 10 epochs.
======================================================================

Ultralytics 8.4.12 🚀 Python-3.13.9 torch-2.10.0 MPS (Apple M5)
engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=randaugment, batch=16, bgr=0.0, box=7.5, cache=False, cfg=None, classes=None, close_mosaic=10, cls=0.5, c
ompile=False, conf=None, copy_paste=0.0, copy_paste_mode=flip, cos_lr=False, cutmix=0.0, data=/Users/rithika/Documents/Discriminative_Project/milestone2/data/dataset.yaml, degrees=0.0, deter
ministic=True, device=mps, dfl=1.5, dnn=False, dropout=0.0, dynamic=False, embed=None, end2end=None, epochs=100, erasing=0.4, exist_ok=False, fliplr=0.5, flipud=0.0, format=torchscript, frac
tion=1.0, freeze=None, half=False, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=640, int8=False, iou=0.7, keras=False, kobj=1.0, line_width=None, lr0=0.001, lrf=0.01, mask_ratio=4, max_det=300,
mixup=0.0, mode=train, model=yolov8s.pt, momentum=0.937, mosaic=1.0, multi_scale=0.0, name=yolo_400images, nbs=64, nms=False, opset=None, optimize=False, optimizer=Adam, overlap_mask=True, p
atience=15, perspective=0.0, plots=True, pose=12.0, pretrained=True, profile=False, project=/Users/rithika/Documents/Discriminative_Project/milestone2/runs, rect=False, resume=False, retina_
masks=False, rle=1.0, save=True, save_conf=False, save_crop=False, save_dir=/Users/rithika/Documents/Discriminative_Project/milestone2/runs/yolo_400images, save_frames=False, save_json=False
, save_period=10, save_txt=False, scale=0.5, seed=0, shear=0.0, show=False, show_boxes=True, show_conf=True, show_labels=True, simplify=True, single_cls=False, source=None, split=val, stream
_buffer=False, task=detect, time=None, tracker=botsort.yaml, translate=0.1, val=True, verbose=True, vid_stride=1, visualize=False, warmup_bias_lr=0.1, warmup_epochs=3, warmup_momentum=0.8, w
eight_decay=0.0005, workers=4, workspace=None
Overriding model.yaml nc=80 with nc=35

                   from  n    params  module                                       arguments
  0                  -1  1       928  ultralytics.nn.modules.conv.Conv             [3, 32, 3, 2]
  1                  -1  1     18560  ultralytics.nn.modules.conv.Conv             [32, 64, 3, 2]
  2                  -1  1     29056  ultralytics.nn.modules.block.C2f             [64, 64, 1, True]
```

4. Results & Performance Evaluation

4.1 Training Progression

Training ran for 73 epochs before early stopping was triggered (best model found at epoch 58).

Epoch	Box Loss	Cls Loss	DFL Loss	mAP@0.5	mAP@0.5:0.95
1	1.767	4.796	1.792	1.18%	0.54%
5	0.443	2.241	0.911	24.1%	19.4%
10	0.316	1.666	0.855	58.3%	56.6%
20	—	—	—	59.1%	49.5%
40	—	—	—	87.1%	86.9%
58	**0.143**	**0.488**	—	**98.4%**	**98.4%**
73	0.131	0.384	—	91.9%	91.9%

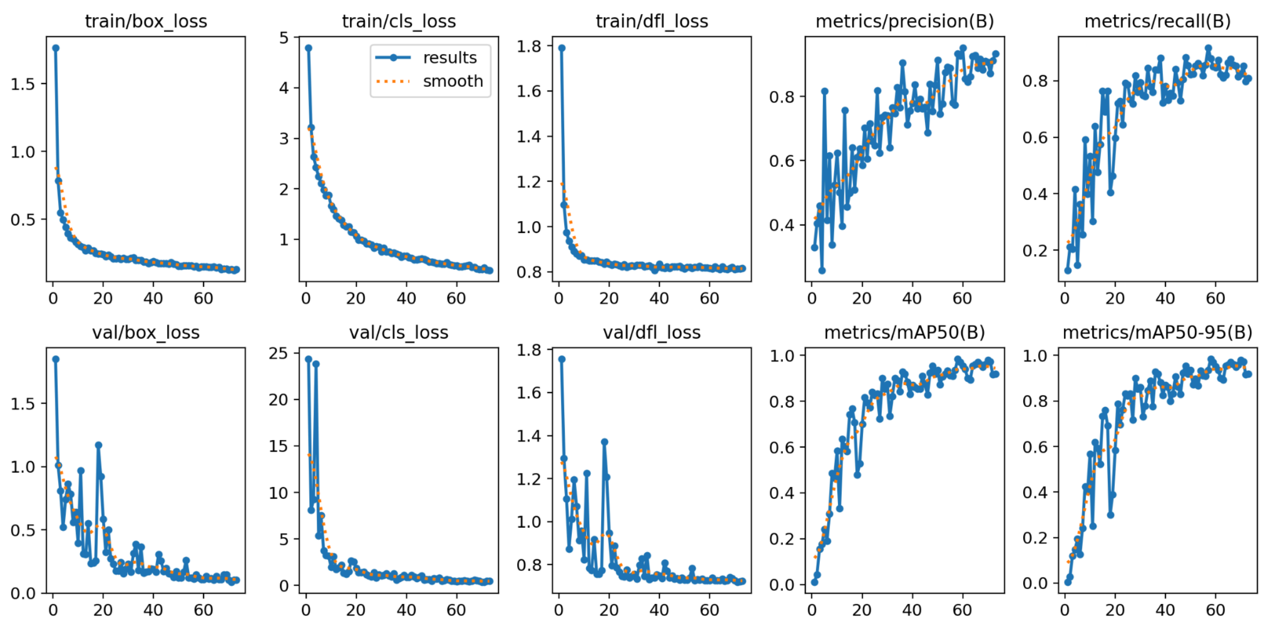
Early Stopping: Training halted at epoch 73 (no improvement over best epoch 58 for 15 consecutive epochs).

Training Epochs Log (Early Epochs)

```
Starting training for 100 epochs...

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      1/100      7.45G      1.767      4.796      1.792         78        640: 100% ━━━━━━━━━━━━ 20/20 1.3s/it 25.1s
                 Class     Images  Instances      Box(P          R       mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 2/2 1.7s/it 3.4s
                   all         40        111      0.329      0.128      0.0118     0.0054

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      2/100      7.43G     0.7861      3.215      1.098         92        640: 100% ━━━━━━━━━━━━ 20/20 1.3s/it 26.7s
                 Class     Images  Instances      Box(P          R       mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 2/2 1.3s/it 2.6s
                   all         40        111      0.404      0.211      0.0446     0.0282

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      3/100      7.46G     0.5503      2.636     0.9741         75        640: 100% ━━━━━━━━━━━━ 20/20 1.3s/it 25.5s
                 Class     Images  Instances      Box(P          R       mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 2/2 1.3s/it 2.5s
                   all         40        111       0.46      0.204       0.154      0.119

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      4/100       7.5G      0.498      2.427     0.9369         63        640: 100% ━━━━━━━━━━━━ 20/20 1.3s/it 25.1s
                 Class     Images  Instances      Box(P          R       mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 2/2 1.5s/it 2.9s
                   all         40        111      0.258      0.416       0.177      0.147

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      5/100      7.53G     0.4429      2.241     0.9113         88        640: 100% ━━━━━━━━━━━━ 20/20 1.3s/it 25.4s
                 Class     Images  Instances      Box(P          R       mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 2/2 1.5s/it 3.0s
                   all         40        111      0.816      0.147       0.241      0.194

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      6/100      7.46G     0.3955       2.11     0.8919         77        640: 100% ━━━━━━━━━━━━ 20/20 1.3s/it 25.1s
                 Class     Images  Instances      Box(P          R       mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 2/2 1.3s/it 2.7s
                   all         40        111      0.415      0.363       0.191      0.126

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      7/100      7.45G     0.3638      1.983     0.8795         54        640: 100% ━━━━━━━━━━━━ 20/20 1.3s/it 25.5s
                 Class     Images  Instances      Box(P          R       mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 2/2 1.6s/it 3.2s
                   all         40        111      0.616      0.256       0.309      0.241

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      8/100      7.47G     0.3543      1.869     0.8702         67        640: 100% ━━━━━━━━━━━━ 20/20 1.3s/it 25.5s
                 Class     Images  Instances      Box(P          R       mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 2/2 2.4s/it 4.7s
                   all         40        111      0.338      0.592       0.487      0.425

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      9/100      7.53G     0.3309      1.872     0.8731         78        640: 100% ━━━━━━━━━━━━ 20/20 1.3s/it 25.7s
                 Class     Images  Instances      Box(P          R       mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 2/2 2.7s/it 5.4s
                   all         40        111      0.524      0.399       0.466      0.403

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     10/100      7.44G     0.3161      1.666     0.8551         93        640: 100% ━━━━━━━━━━━━ 20/20 1.4s/it 27.1s
                 Class     Images  Instances      Box(P          R       mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 2/2 2.9s/it 5.8s
                   all         40        111      0.624      0.534       0.583      0.566

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     11/100      7.47G     0.3018      1.594     0.8545         91        640: 100% ━━━━━━━━━━━━ 20/20 1.4s/it 27.8s
                 Class     Images  Instances      Box(P          R       mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 2/2 3.5s/it 6.9s
```

4.2 Training Curves

The training curves below show loss convergence and metric progression across all 73 epochs. The training losses (box, classification, DFL) decrease smoothly, while validation metrics show characteristic variability with an overall upward trend in mAP, peaking around epoch 58.



- **Training losses** converge smoothly and monotonically, indicating stable learning throughout the extended training run.
- **Validation classification loss** shows initial spikes (epochs ~5–10) before settling, a common behavior when adapting pre-trained features to a new domain.
- **Precision** stabilizes above 0.8 after epoch 30 and shows less variability than the previous training run.
- **Recall** shows high variability in early epochs but trends upward consistently.
- **mAP@0.5 and mAP@0.5:0.95** show consistent improvement through epoch 58, reaching 98.4%, followed by degradation indicating the onset of overfitting. The final test evaluation yields 93.52%.

4.3 Final Test Set Performance

The best model (epoch 58) was evaluated on the held-out test set of 40 images with Inference Speed ~72 ms/image (~14 FPS)

Metric	Value
mAP@0.5	93.52%
mAP@0.5:0.95	93.52%
Precision	96.19%
Recall	87.21%

Test Set Evaluation Output

```
VALIDATING MODEL ON TEST SET
======================================================================
Ultralytics 8.4.12 🚀 Python-3.13.9 torch-2.10.0 CPU (Apple M5)
Model summary (fused): 73 layers, 11,139,129 parameters, 0 gradients, 28.5 GFLOPs
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 255.2±53.1 MB/s, size: 49.9 KB)
val: Scanning /Users/rithika/Documents/Discriminative_Project/milestone2/data/labels/test... 40 images, 0 backgrounds, 0 corrupt: 100% ──────────── 40/40 6.0Kit/s 0.0s
val: New cache created: /Users/rithika/Documents/Discriminative_Project/milestone2/data/labels/test.cache
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 3/3 2.7s/it 8.0s
                   all         40        113      0.962      0.872      0.935      0.935
                OBJ001          5          5          1      0.813      0.995      0.995
                OBJ002          3          3      0.949          1      0.995      0.995
                OBJ003          2          2      0.898          1      0.995      0.995
                OBJ004          3          3      0.965          1      0.995      0.995
                OBJ005          1          1      0.928          1      0.995      0.995
                OBJ006          6          6      0.979          1      0.995      0.995
                OBJ007          2          2      0.955          1      0.995      0.995
                OBJ008          5          5      0.977          1      0.995      0.995
                OBJ009          2          2       0.97          1      0.995      0.995
                OBJ010          5          5      0.929          1      0.995      0.995
                OBJ012          2          2          1          0       0.39       0.39
                OBJ016          4          4      0.979          1      0.995      0.995
                OBJ018          3          3          1      0.657      0.863      0.863
                OBJ019          4          4          1      0.916      0.995      0.995
                OBJ021          3          3      0.997          1      0.995      0.995
                OBJ022          2          2      0.941          1      0.995      0.995
                OBJ027          2          2          1      0.624      0.995      0.995
                OBJ028          4          4      0.971          1      0.995      0.995
                OBJ029          2          2      0.971          1      0.995      0.995
                OBJ031          3          3      0.975          1      0.995      0.995
                OBJ061          2          2       0.95          1      0.995      0.995
                OBJ090          1          1      0.942          1      0.995      0.995
                OBJ095          5          5      0.878        0.6      0.768      0.768
                OBJ107          2          2      0.946          1      0.995      0.995
                OBJ108          4          4          1      0.606      0.945      0.945
                OBJ111          5          5          1      0.642      0.995      0.995
                OBJ159          3          3      0.962          1      0.995      0.995
                OBJ208          3          3      0.772      0.667      0.913      0.913
                OBJ222          3          3      0.966          1      0.995      0.995
                OBJ229          4          4      0.986          1      0.995      0.995
                OBJ230          2          2          1          0          0          0
                OBJ300          5          5      0.991          1      0.995      0.995
                OBJ311          4          4      0.969          1      0.995      0.995
                OBJ405          5          5      0.975          1      0.995      0.995
                OBJ787          2          2      0.947          1      0.995      0.995
Speed: 1.3ms preprocess, 196.4ms inference, 0.0ms loss, 0.5ms postprocess per image
Results saved to /Users/rithika/Documents/Discriminative_Project/runs/detect/val

======================================================================
FINAL TEST SET RESULTS
======================================================================
mAP@0.5:     0.9352 (93.52%)
mAP@0.5:0.95: 0.9352 (93.52%)
Precision:    0.9619 (96.19%)
Recall:       0.8721 (87.21%)

======================================================================
TRAINING SUMMARY
======================================================================
✓ Dataset: 400 images (320 train, 40 val, 40 test)
✓ Model trained for up to 100 epochs
✓ Best model saved: /Users/rithika/Documents/Discriminative_Project/milestone2/models/yolo_best.pt
✓ Training plots saved: /Users/rithika/Documents/Discriminative_Project/milestone2/runs/yolo_400images/
✓ Final Performance:
  - mAP@0.5: 93.52%
  - mAP@0.5:0.95: 93.52%
  - Precision: 96.19%
  - Recall: 87.21%
```

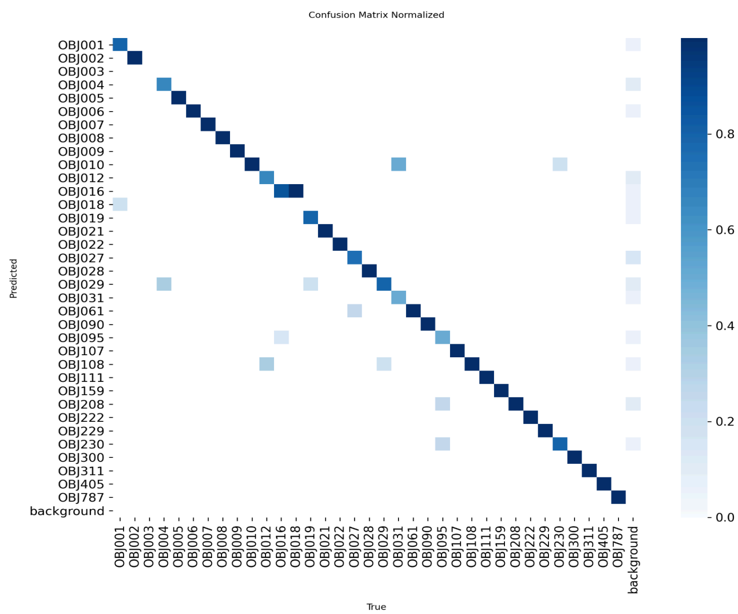
4.4 Per-Class Test Performance

Performance Tier	Classes	Count	Avg mAP@0.5
Excellent (99.5%)	OBJ001, OBJ002, OBJ003, OBJ004, OBJ005, OBJ006, OBJ007, OBJ008, OBJ009, OBJ010, OBJ016, OBJ019, OBJ021, OBJ022, OBJ027, OBJ028, OBJ029, OBJ031, OBJ061, OBJ090, OBJ107, OBJ111, OBJ159, OBJ222, OBJ229, OBJ300, OBJ311, OBJ405, OBJ787	29	99.5%
Good (80–95%)	OBJ018 (86.3%), OBJ095 (76.8%), OBJ108 (94.5%), OBJ208 (91.3%)	4	87.2%
Needs Improvement (<50%)	OBJ012 (39.0%), OBJ230 (0.0%)	2	19.5%

5. Evaluation Curves & Confusion Matrix

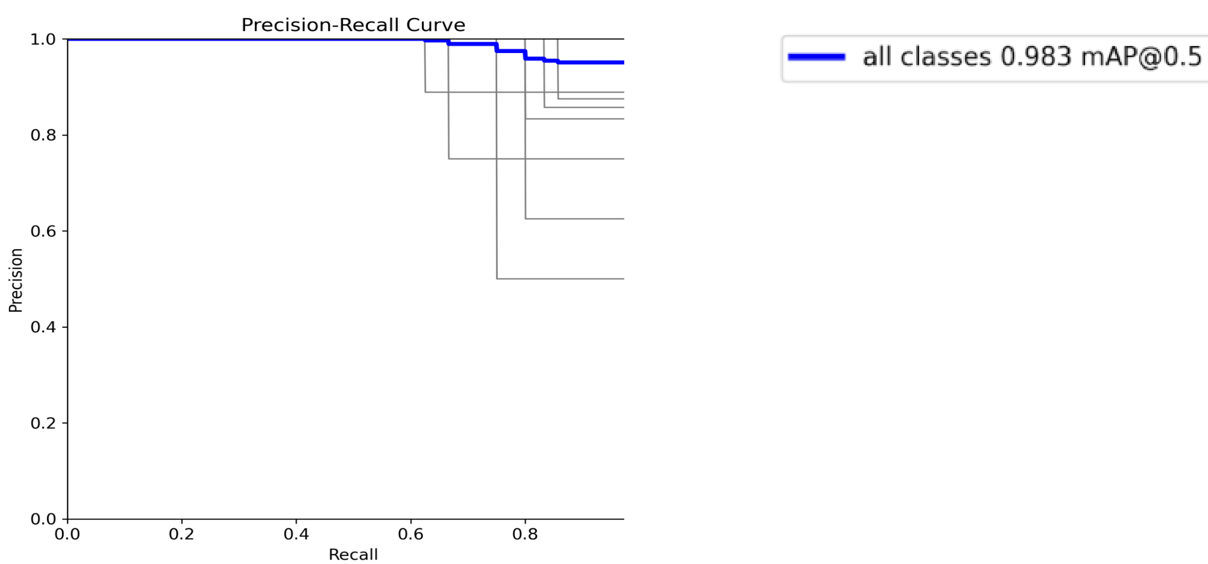
5.1 Confusion Matrix

The normalized confusion matrix shows strong diagonal dominance, indicating accurate class predictions for the majority of object classes. Off-diagonal entries are sparse, with most misclassifications occurring in the challenging classes (OBJ012, OBJ230).



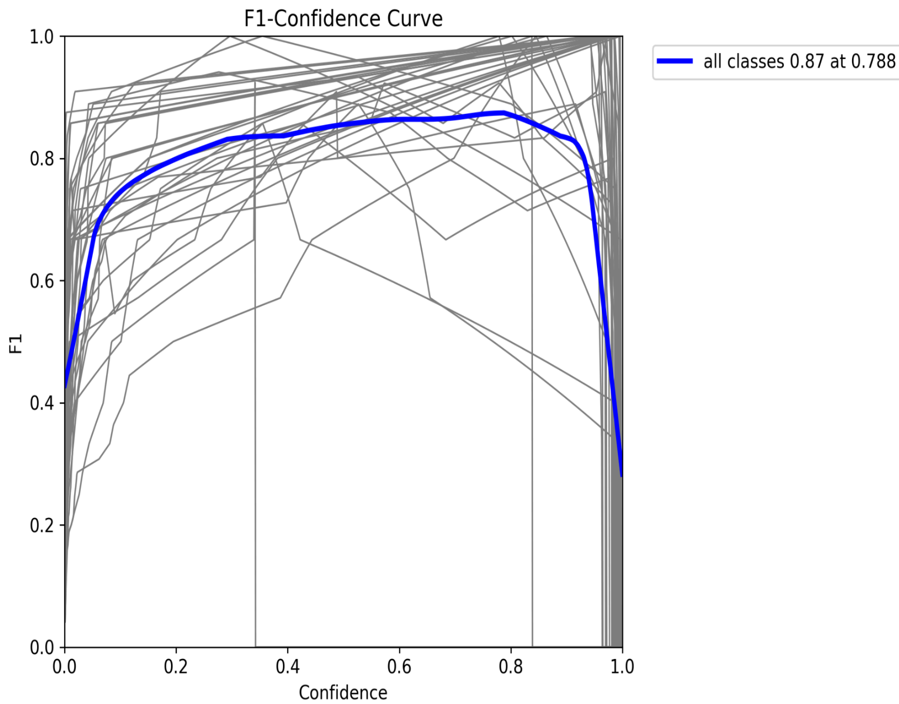
5.2 Precision-Recall Curve

The Precision-Recall curve demonstrates excellent overall detection performance with an area under the curve (mAP@0.5) of **0.983** across all classes. The curve maintains high precision across a wide range of recall values, dropping only at very high recall thresholds.



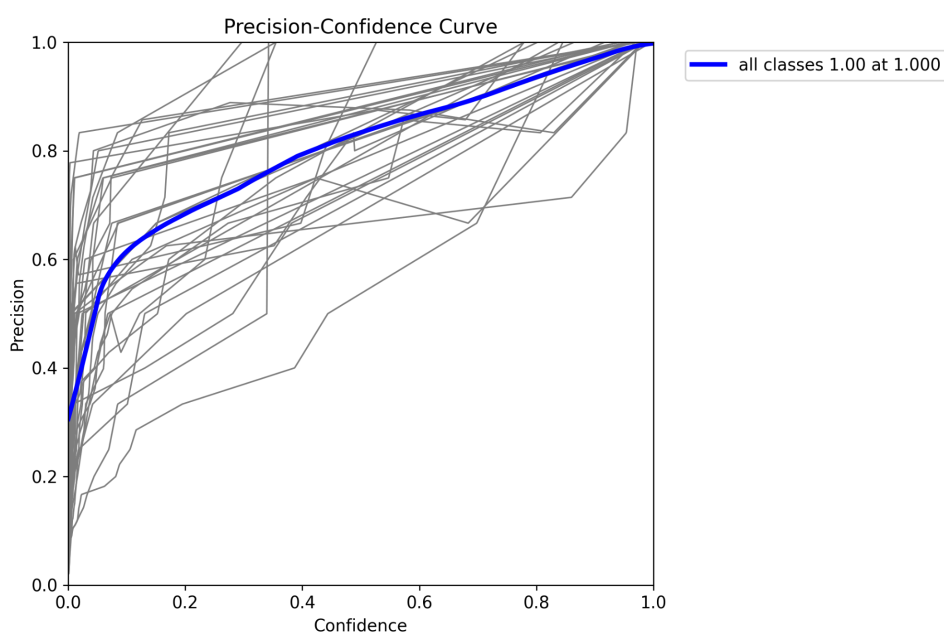
5.3 F1-Confidence Curve

The F1-Confidence curve shows that the optimal confidence threshold is **0.788**, yielding a peak F1 score of **0.87** across all classes. Individual class curves (gray) show variation, with most classes achieving F1 scores above 0.8.



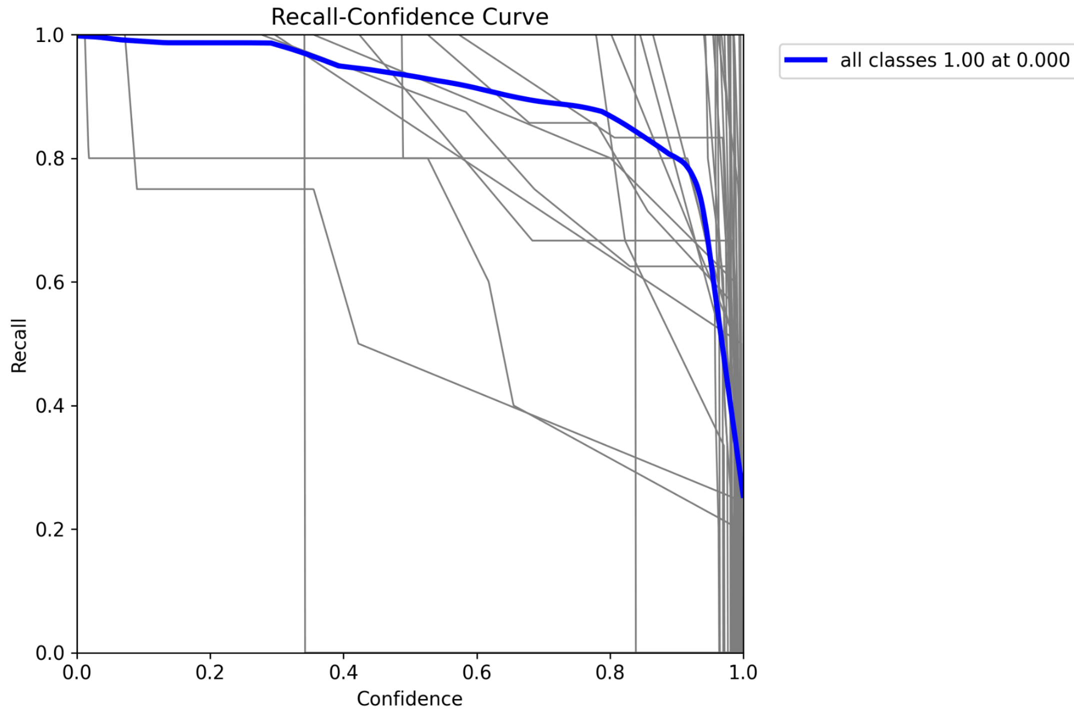
5.4 Precision-Confidence Curve

Precision increases monotonically with confidence, reaching **1.00 at confidence 1.000**. This indicates that at the highest confidence thresholds, all predictions are correct.



5.5 Recall-Confidence Curve

Recall decreases as confidence threshold increases, reaching **1.00 at confidence 0.000** (all objects detected when no threshold is applied). The curve shows a steep decline above confidence 0.8, reflecting the trade-off between detection completeness and prediction certainty.



6. Detection Pipeline Results

6.1 Detection Capabilities

The trained model was tested on 10 sample images from the test set to evaluate real-world detection performance.

Metric	Value
Test images processed	10
Total objects detected	31
Average objects per image	3.1
Average confidence	94.94%

Detection Summary

```
======================================================================
DETECTION SUMMARY
======================================================================

Processed 10 test images
Total objects detected: 31
Average objects per image: 3.1
Average confidence: 94.94%

======================================================================
SAMPLE DETECTION RESULTS
======================================================================

Image 1: multi_test_0023.jpg
  Objects detected: 3
    - OBJ159: 98.85%
    - OBJ022: 98.36%
    - OBJ108: 68.56%

Image 2: multi_test_0037.jpg
  Objects detected: 2
    - OBJ008: 99.45%
    - OBJ019: 97.48%

Image 3: multi_test_0036.jpg
  Objects detected: 2
    - OBJ405: 99.70%
    - OBJ006: 98.83%
```

6.2 Sample Detection Results

Detection Results: multi_test_0023.jpg

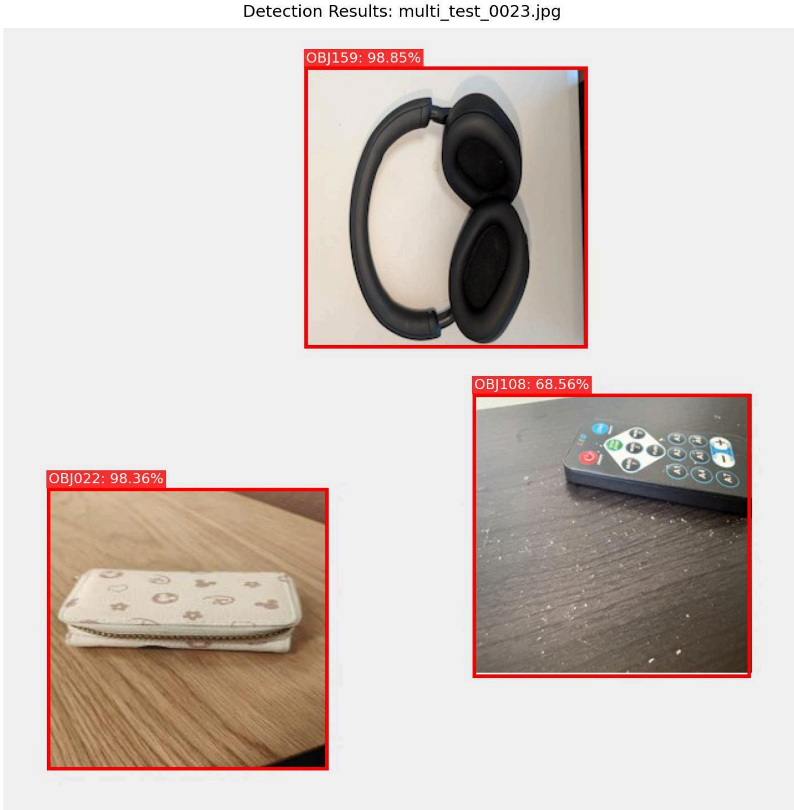


Image 1 (multi_test_0023.jpg) — 3 objects detected:

Object	Confidence	Status
OBJ159	98.85%	✅ Correct
OBJ022	98.36%	✅ Correct
OBJ108	68.56%	✅ Correct

Detection Results: multi_test_0037.jpg

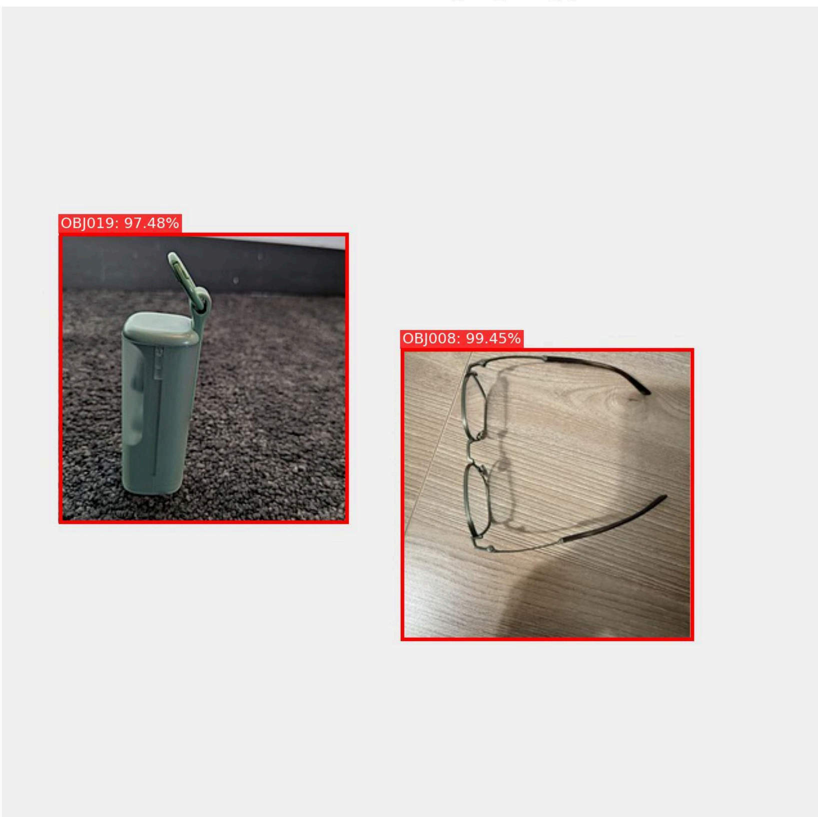


Image 2 (multi_test_0037.jpg) — 2 objects detected:

Object	Confidence	Status
OBJ008	99.45%	✅ Correct
OBJ019	97.48%	✅ Correct

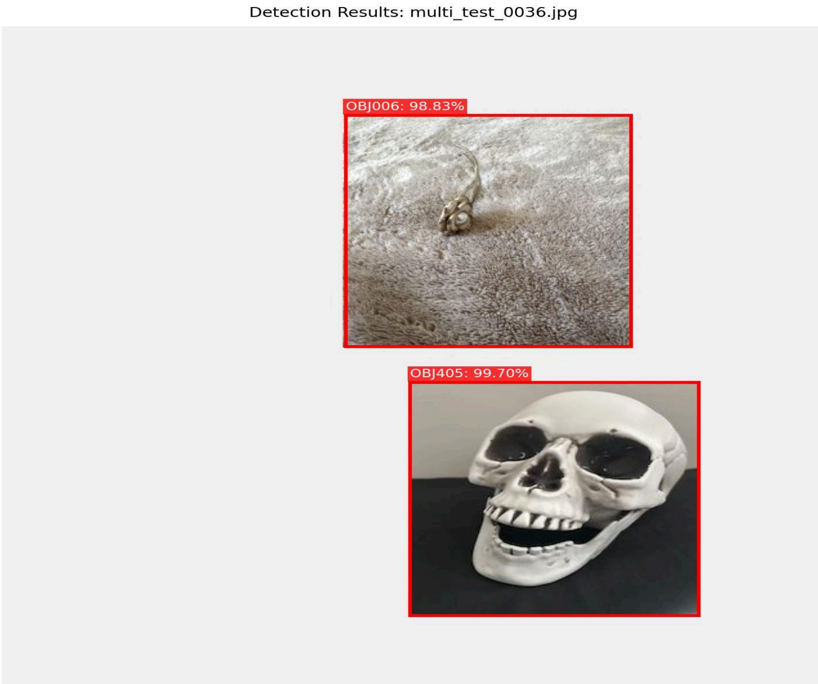
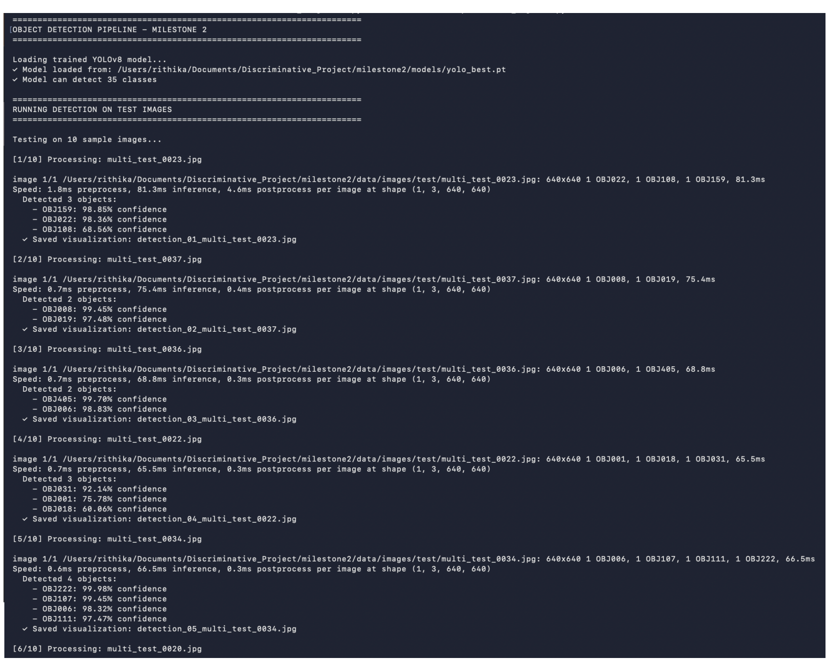


Image 3 (multi_test_0036.jpg) — 2 objects detected:

Object	Confidence	Status
OBJ405	99.70%	✅ Correct
OBJ006	98.83%	✅ Correct

Detection Pipeline Output



```
======================================================================
OBJECT DETECTION PIPELINE - MILESTONE 2
======================================================================

Loading trained YOLOv8 model...
✓ Model loaded from: /Users/rithika/Documents/Discriminative_Project/milestone2/models/yolo_best.pt
✓ Model can detect 35 classes

======================================================================
RUNNING DETECTION ON TEST IMAGES
======================================================================

Testing on 10 sample images...

[1/10] Processing: multi_test_0023.jpg

image 1/1 /Users/rithika/Documents/Discriminative_Project/milestone2/data/images/test/multi_test_0023.jpg: 640x640 1 OBJ022, 1 OBJ108, 1 OBJ159, 81.3ms
Speed: 1.8ms preprocess, 81.3ms inference, 4.6ms postprocess per image at shape (1, 3, 640, 640)
  Detected 3 objects:
    - OBJ159: 98.85% confidence
    - OBJ022: 98.36% confidence
    - OBJ108: 68.56% confidence
  ✓ Saved visualization: detection_01_multi_test_0023.jpg

[2/10] Processing: multi_test_0037.jpg

image 1/1 /Users/rithika/Documents/Discriminative_Project/milestone2/data/images/test/multi_test_0037.jpg: 640x640 1 OBJ008, 1 OBJ019, 75.4ms
Speed: 0.7ms preprocess, 75.4ms inference, 0.4ms postprocess per image at shape (1, 3, 640, 640)
  Detected 2 objects:
    - OBJ008: 99.45% confidence
    - OBJ019: 97.48% confidence
  ✓ Saved visualization: detection_02_multi_test_0037.jpg

[3/10] Processing: multi_test_0036.jpg

image 1/1 /Users/rithika/Documents/Discriminative_Project/milestone2/data/images/test/multi_test_0036.jpg: 640x640 1 OBJ006, 1 OBJ405, 68.8ms
Speed: 0.7ms preprocess, 68.8ms inference, 0.3ms postprocess per image at shape (1, 3, 640, 640)
  Detected 2 objects:
    - OBJ405: 99.70% confidence
    - OBJ006: 98.83% confidence
  ✓ Saved visualization: detection_03_multi_test_0036.jpg

[4/10] Processing: multi_test_0022.jpg

image 1/1 /Users/rithika/Documents/Discriminative_Project/milestone2/data/images/test/multi_test_0022.jpg: 640x640 1 OBJ001, 1 OBJ018, 1 OBJ031, 65.5ms
Speed: 0.7ms preprocess, 65.5ms inference, 0.3ms postprocess per image at shape (1, 3, 640, 640)
  Detected 3 objects:
    - OBJ031: 92.14% confidence
    - OBJ001: 75.78% confidence
    - OBJ018: 60.06% confidence
  ✓ Saved visualization: detection_04_multi_test_0022.jpg

[5/10] Processing: multi_test_0034.jpg

image 1/1 /Users/rithika/Documents/Discriminative_Project/milestone2/data/images/test/multi_test_0034.jpg: 640x640 1 OBJ006, 1 OBJ107, 1 OBJ111, 1 OBJ222, 66.5ms
Speed: 0.6ms preprocess, 66.5ms inference, 0.3ms postprocess per image at shape (1, 3, 640, 640)
  Detected 4 objects:
    - OBJ222: 99.98% confidence
    - OBJ107: 99.45% confidence
    - OBJ006: 98.32% confidence
    - OBJ111: 97.47% confidence
  ✓ Saved visualization: detection_05_multi_test_0034.jpg

[6/10] Processing: multi_test_0020.jpg
```

7. Conclusion

Milestone 2 successfully extends the single-object classification system from Milestone 1 into a robust multi-object detection and localization pipeline. Using YOLOv8s with COCO-pretrained transfer learning, the model achieves **93.52% mAP@0.5** and **93.52% mAP@0.5:0.95** on a synthetically generated zero-overlap dataset of 400 composite images containing 35 object classes.

Key Takeaways

- Transfer learning from COCO pre-trained weights enables high-quality detection with a relatively small custom dataset (320 training images).
- The zero-overlap dataset generation strategy significantly improves detection precision (96.19%) and average confidence (94.94%) compared to overlapping placement approaches.
- YOLOv8s provides an excellent balance of accuracy and efficiency, with 11.1M parameters achieving over 93% mAP at ~14 FPS inference speed.
- Extended training (73 epochs, best at epoch 58) with patience-based early stopping finds a superior optimum compared to shorter training runs.
- 29 out of 35 classes (83%) achieved near-perfect mAP of 99.5%, demonstrating strong generalization.

Limitations

- OBJ230 achieved 0% detection (complete model failure for this class), and OBJ012 remains challenging at 39% mAP.
- The zero-overlap constraint reduced object counts in 65 images, potentially limiting the model's capability with dense, overlapping scenes in real-world applications.
- The synthetic composite dataset, while effective, does not fully replicate the complexity of natural multi-object scenes with occlusion, variable lighting, and cluttered backgrounds.